Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3291-WE01

Title:

MEng/BEng: Thermodynamics and Fluid Mechanics 3

Time Allowed:	2 hours	
Additional Material provided:	No	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.

Visiting Students may use dictionaries: Yes

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Q1

(a) Sketch an enthalpy-entropy diagram of a generic gas turbine compressor stage and show that the stagnation pressure at the stage exit p_{03} can be expressed in terms of the stage inlet stagnation pressure p_{01} in the following way:

$$p_{03} = p_{01}\left(1 + \frac{w_x \eta_{tt}}{C_p T_{01}}\right)^{\frac{\gamma}{\gamma-1}},$$

where the other variables and constants have their usual meaning.

[30%]

The flowrate through a standard Francis turbine is 10 m^3/s. The stator blade height is 200 mm, and the diameter of the machine at rotor entry is 1400 mm. The blockage in the stator can be assumed to be 5%. The turbine rotates at 300 rpm and the stator exit angle is 80°. The rotor exit velocity angle is 10° and the mean exit diameter is 1240mm. The effective blade height at rotor exit is 400mm, where the blockage is assumed to be 8%.

(b) Calculate the radial velocity at the inlet to the stator.

[15%]

(c) Sketch the cascade view and velocity triangles for rotor inlet and exit. Determine the rotor relative inlet and exit angles.

[40%]

(d) Calculate the power output of the turbine.

[15%]

continued

Q2

(a) What are the most likely outcomes of neutron interaction with matter? Describe each outcome, provide its reaction equation and explain how the probability of each outcome can be quantified.

[10%]

(b) Give pros and cons of nuclear fusion power over those of nuclear fission.

[10%]

(c) Using data in Table Q2 calculate the Q-value of the alpha decay of Beryllium-8.

[10%]

Element	Symbol	Atomic number (Z)	Atomic mass(u)
Hydrogen	^{2}H	1	2.014102
Hydrogen	^{3}H	1	3.016049
Helium	^{3}He	2	3.016029
Helium	^{4}He	2	4.002602
Helium	^{5}He	2	5.012224
Beryllium	^{8}Be	4	8.005305
Beryllium	^{9}Be	4	9.012182
Uranium	^{234}U	92	234.040952
Uranium	^{235}U	92	235.043930
Uranium	^{236}U	92	236.045569

Table Q2

(d) A nuclear fission reactor of volume 1.4 m^3 uses U-235 isotope fuel and thermal neutrons with E_k = 0.025 eV. At this kinetic energy the microscopic fission cross-section of U-235 is 582 barns. The fission of one nucleus of U-235 releases approximately 200 MeV of energy, density of U-235 is 18900 kg/m^3, Avogadro number is 6.022×10^{26} atoms/kmol and the relations between atomic units and SI units are: 1 MeV $= 1.602 \times 10^{-13}$ J and 1 u $= 1.6605 \times 10^{-27}$ kg. Using data provided above and in Table Q2 calculate the macroscopic fission cross-section of U-235.

[10%]

(e) What neutron flux is required to generate 60 MW of power using the reactor in part (d)?

[10%]

(f) How much fuel (in kg) is necessary for the reactor in part (d) to operate at full power for 24 hours?

[15%]

(g) Describe the principal differences between the steam cycles of typical PWR and AGR nuclear power plants.

[20%]

continued

(h) What roles are the following types of large electric power plants likely to play in the UK's low/zero carbon future energy plans?

i) Combined Cycle Power Plants

ii) Coal Fired Power Plants

iii) Nuclear Power Plants

[**15%**]

Q3

At design stage, the engine intake of a supersonic flying vehicle is simplified as a two-dimensional geometry to be tested in a supersonic wind tunnel; see Fig. Q3. The model has a width (into the paper/screen) of 1 m. As shown, the incoming flow is directed horizontally and has a Mach number of 3.0, a stagnation temperature of $T_0 = 500$ K and a stagnation pressure of $p_0 = 3$ bar. The wedge surface AB is at an angle of 20° to the horizontal, while the surface BD is horizontal and along which there is one side of a duct CD. The other side of the duct, C'D', of the same length, lies parallel to it and a constant height of 0.3 m away; both sides of the duct have an average friction factor of $f = 0.005$. The section BC can be considered frictionless. You may also assume that the locations A, B and C are far enough apart that there is no interaction of any flow features which arise in their vicinity.

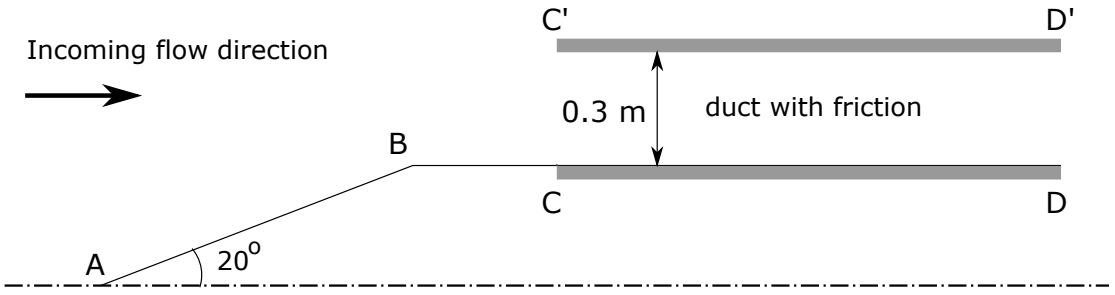


Fig. Q3: Not to scale

(a) Sketch and label the aerodynamic flow features that develop for the given geometry. Provide basic justifications without detailed calculations.

[10%]

(b) Calculate the mass flow rate $\dot{m}$ entering the duct (CC') and the length of the duct CD=L_{CD}, such that the flow is choked at DD'.

[55%]

(c) For the same duct length L_{CD}, the exit condition at DD' is now changed such that there is a normal shock locating at $\frac{1}{3}L_{CD}$ along the duct (from the CC' end). Calculate the exit Mach number at DD'.

[35%]

END